Year 9

REVISION PACK

Unit 3

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Unit 3 · Arithmetic sequences (Lesson 3.1)

Q	Working	Answer
nth term and testing a number		
1.1	**(a)** $\text{difference} = 4$ $4n:\ 4, 8, 12, 16$ $\text{each term is 1 more than } 4n$ $n\text{th term} = 4n + 1$ **(b)** $4n + 1 = 81$ $4n = 80$ $n = 20$	**(a)** $4n + 1$ **(b) Yes, it is the 20th term**
1.2	**(a)** $\text{difference} = 6$ $6n:\ 6, 12, 18, 24$ $\text{each term is 4 less than } 6n$ $n\text{th term} = 6n - 4$ **(b)** $6n - 4 = 75$ $6n = 79$ $n = 13.17\ldots$	**(a)** $6n - 4$ **(b) No,** $n = 79 \div 6$ **is not a whole number**
1.3	**(a)** $\text{difference} = -3$ $-3n:\ -3, -6, -9, -12$ $\text{each term is 23 more than } -3n$ $n\text{th term} = -3n + 23$ **(b)** $-3n + 23 = 2$ $-3n = -21$ $n = 7$	**(a)** $-3n + 23$ **(b) Yes, it is the 7th term**

1.4	**(a)** $\text{difference} = 7$ $7n:\ 7, 14, 21, 28$ $\text{each term is } 4 \text{ less than } 7n$ $n\text{th term} = 7n - 4$ **(b)** $7n - 4 = 150$ $7n = 154$ $n = 22$	**(a)** $7n - 4$ **(b) Yes, it is the 22th term**
1.5	**(a)** $\text{difference} = 5$ $5n:\ 5, 10, 15, 20$ $\text{each term is } 6 \text{ more than } 5n$ $n\text{th term} = 5n + 6$ **(b)** $5n + 6 = 90$ $5n = 84$ $n = 16.8\ldots$	**(a)** $5n + 6$ **(b) No,** $n = 84 \div 5$ **is not a whole number**
1.6	**(a)** $\text{difference} = -6$ $-6n:\ -6, -12, -18, -24$ $\text{each term is } 46 \text{ more than } -6n$ $n\text{th term} = -6n + 46$ **(b)** $-6n + 46 = 4$ $-6n = -42$ $n = 7$	**(a)** $-6n + 46$ **(b) Yes, it is the 7th term**

Unit 3 · Quadratic sequences (Lesson 3.1)

Q	Working	Answer
next term and nth term		

2.1	**(a)** 1st differences: $3, 5, 7, 9$ 2nd difference: 2 next difference $= 9 + 2 = 11$ $26 + 11 = 37$ **(b)** 2nd difference: 2 $2 \div 2 = 1$ n^2 term subtract n^2 : $1, 1, 1, 1, 1$ 1 nth term $= n^2 + 1$	**(a) 37** **(b)** $n^2 + 1$
2.2	**(a)** 1st differences: $6, 8, 10, 12$ 2nd difference: 2 next difference $= 12 + 2 = 14$ $39 + 14 = 53$ **(b)** 2nd difference: 2 $2 \div 2 = 1$ n^2 term subtract n^2 : $2, 5, 8, 11, 14$ $3n - 1$ nth term $= n^2 + 3n - 1$	**(a) 53** **(b)** $n^2 + 3n - 1$
2.3	**(a)** 1st differences: $5, 7, 9, 11$ 2nd difference: 2 next difference $= 11 + 2 = 13$ $38 + 13 = 51$ **(b)** 2nd difference: 2 $2 \div 2 = 1$ n^2 term subtract n^2 : $5, 7, 9, 11, 13$ $2n + 3$ nth term $= n^2 + 2n + 3$	**(a) 51** **(b)** $n^2 + 2n + 3$

2.4	**(a)** 1st differences: $7, 9, 11, 13$ 2nd difference: 2 next difference $= 13 + 2 = 15$ $45 + 15 = 60$ **(b)** 2nd difference: 2 $2 \div 2 = 1$ n^2 term subtract n^2 : $4, 8, 12, 16, 20$ $4n$ nth term $= n^2 + 4n$	**(a) 60** **(b)** $n^2 + 4n$
2.5	**(a)** 1st differences: $8, 10, 12, 14$ 2nd difference: 2 next difference $= 14 + 2 = 16$ $50 + 16 = 66$ **(b)** 2nd difference: 2 $2 \div 2 = 1$ n^2 term subtract n^2 : $5, 10, 15, 20, 25$ $5n$ nth term $= n^2 + 5n$	**(a) 66** **(b)** $n^2 + 5n$
2.6	**(a)** 1st differences: $2, 4, 6, 8$ 2nd difference: 2 next difference $= 8 + 2 = 10$ $23 + 10 = 33$ **(b)** 2nd difference: 2 $2 \div 2 = 1$ n^2 term subtract n^2 : $2, 1, 0, -1, -2$ $-n + 3$ nth term $= n^2 - n + 3$	**(a) 33** **(b)** $n^2 - n + 3$

Unit 3 · Geometric sequences (Lesson 3.2)

Q	Working	Answer
term-to-term rule and a later term		

3.1	**(a)** $6 \div 2 = 3$ **(b)** terms: $2, 6, 18, 54, 162, 486$ $6\text{th term} = 162 \times 3 = 486$	**(a) Multiply by 3** **(b) 486**
3.2	**(a)** $10 \div 5 = 2$ **(b)** terms: $5, 10, 20, 40, 80, 160, 320, 640$ $8\text{th term} = 320 \times 2 = 640$	**(a) Multiply by 2** **(b) 640**
3.3	**(a)** $4 \div 1 = 4$ **(b)** terms: $1, 4, 16, 64, 256$ $5\text{th term} = 64 \times 4 = 256$	**(a) Multiply by 4** **(b) 256**
3.4	**(a)** $12 \div 4 = 3$ **(b)** terms: $4, 12, 36, 108, 324$ $5\text{th term} = 108 \times 3 = 324$	**(a) Multiply by 3** **(b) 324**
3.5	**(a)** $20 \div 10 = 2$ **(b)** terms: $10, 20, 40, 80, 160, 320$ $6\text{th term} = 160 \times 2 = 320$	**(a) Multiply by 2** **(b) 320**
3.6	**(a)** $3 \div 1 = 3$ **(b)** terms: $1, 3, 9, 27, 81, 243, 729$ $7\text{th term} = 243 \times 3 = 729$	**(a) Multiply by 3** **(b) 729**

Unit 3 · Expanding brackets (Lesson 3.3)

Q	Working	Answer
expand and simplify		
4.1	**(a)** $3x + 12 - 2x - 2$ $= x + 10$ **(b)** $x^2 + 6x + 2x + 12$ $= x^2 + 8x + 12$ **(c)** $(x + 5)(x + 5)$ $= x^2 + 5x + 5x + 25$ $= x^2 + 10x + 25$	**(a)** $x + 10$ **(b)** $x^2 + 8x + 12$ **(c)** $x^2 + 10x + 25$

4.2	**(a)** $5x - 10 - 3x + 12$ $= 2x + 2$ **(b)** $x^2 - 4x + 5x - 20$ $= x^2 + x - 20$ **(c)** $(x - 3)(x - 3)$ $= x^2 - 3x - 3x + 9$ $= x^2 - 6x + 9$	**(a)** $2x + 2$ **(b)** $x^2 + x - 20$ **(c)** $x^2 - 6x + 9$
4.3	**(a)** $6x - 2 - 4x + 12$ $= 2x + 10$ **(b)** $x^2 - 7x - 3x + 21$ $= x^2 - 10x + 21$ **(c)** $(3x + 2)(3x + 2)$ $= 9x^2 + 6x + 6x + 4$ $= 9x^2 + 12x + 4$	**(a)** $2x + 10$ **(b)** $x^2 - 10x + 21$ **(c)** $9x^2 + 12x + 4$
4.4	**(a)** $6x + 6 - 2x - 10$ $= 4x - 4$ **(b)** $x^2 + 7x + 3x + 21$ $= x^2 + 10x + 21$ **(c)** $(x + 4)(x + 4)$ $= x^2 + 4x + 4x + 16$ $= x^2 + 8x + 16$	**(a)** $4x - 4$ **(b)** $x^2 + 10x + 21$ **(c)** $x^2 + 8x + 16$
4.5	**(a)** $6x - 3 - 2x + 6$ $= 4x + 3$ **(b)** $x^2 + 6x - 2x - 12$ $= x^2 + 4x - 12$ **(c)** $(x - 6)(x - 6)$ $= x^2 - 6x - 6x + 36$ $= x^2 - 12x + 36$	**(a)** $4x + 3$ **(b)** $x^2 + 4x - 12$ **(c)** $x^2 - 12x + 36$

4.6	**(a)** $8x + 12 - 3x + 15$ $= 5x + 27$ **(b)** $x^2 - x - 5x + 5$ $= x^2 - 6x + 5$ **(c)** $(2x - 3)(2x - 3)$ $= 4x^2 - 6x - 6x + 9$ $= 4x^2 - 12x + 9$	**(a)** $5x + 27$ **(b)** $x^2 - 6x + 5$ **(c)** $4x^2 - 12x + 9$

Unit 3 · Factorising (Lesson 3.4)

Q	Working	Answer
factorise fully		
5.1	**(a)** $\text{HCF} = 3x$ **(b)** $3 \times 4 = 12, \quad 3 + 4 = 7$ **(c)** $25 = 5^2$	**(a)** $3x(2x + 3)$ **(b)** $(x + 3)(x + 4)$ **(c)** $(x + 5)(x - 5)$
5.2	**(a)** $\text{HCF} = 4x$ **(b)** $2 \times 6 = 12, \quad 2 + 6 = 8$ **(c)** $64 = 8^2$	**(a)** $4x(2x - 3)$ **(b)** $(x + 2)(x + 6)$ **(c)** $(x + 8)(x - 8)$
5.3	**(a)** $\text{HCF} = 7x$ **(b)** $5 \times (-2) = -10, \quad 5 + (-2) = 3$ **(c)** $100 = 10^2$	**(a)** $7x(2x - 3)$ **(b)** $(x + 5)(x - 2)$ **(c)** $(x + 10)(x - 10)$
5.4	**(a)** $\text{HCF} = 3x$ **(b)** $1 \times 6 = 6, \quad 1 + 6 = 7$ **(c)** $36 = 6^2$	**(a)** $3x(3x + 4)$ **(b)** $(x + 1)(x + 6)$ **(c)** $(x + 6)(x - 6)$
5.5	**(a)** $\text{HCF} = 5x$ **(b)** $3 \times (-5) = -15, \quad 3 + (-5) = -2$ **(c)** $121 = 11^2$	**(a)** $5x(3x - 2)$ **(b)** $(x + 3)(x - 5)$ **(c)** $(x + 11)(x - 11)$

5.6	**(a)** $\text{HCF} = 4x$ **(b)** $-3 \times (-4) = 12, \quad -3 + (-4) = -7$ **(c)** $16 = 4^2$	**(a)** $4x(5x + 2)$ **(b)** $(x - 3)(x - 4)$ **(c)** $(x + 4)(x - 4)$

Unit 3 · Solving quadratic equations (Lesson 3.5)

Q	Working	Answer
solve the equations		
6.1	**(a)** $(x - 2)(x - 3) = 0$ $x = 2$ or $x = 3$ **(b)** $x^2 - 4x = 0$ $x(x - 4) = 0$ $x = 0$ or $x = 4$	**(a)** $x = 2$ **or** $x = 3$ **(b)** $x = 0$ **or** $x = 4$
6.2	**(a)** $(x - 6)(x + 1) = 0$ $x = 6$ or $x = -1$ **(b)** $x^2 - 7x = 0$ $x(x - 7) = 0$ $x = 0$ or $x = 7$	**(a)** $x = 6$ **or** $x = -1$ **(b)** $x = 0$ **or** $x = 7$
6.3	**(a)** $(x + 4)(x - 3) = 0$ $x = -4$ or $x = 3$ **(b)** $x^2 - 10x = 0$ $x(x - 10) = 0$ $x = 0$ or $x = 10$	**(a)** $x = -4$ **or** $x = 3$ **(b)** $x = 0$ **or** $x = 10$
6.4	**(a)** $(x - 5)(x + 2) = 0$ $x = 5$ or $x = -2$ **(b)** $x^2 - 3x = 0$ $x(x - 3) = 0$ $x = 0$ or $x = 3$	**(a)** $x = 5$ **or** $x = -2$ **(b)** $x = 0$ **or** $x = 3$

6.5	**(a)** $(x+1)(x+6)=0$ $x=-1$ or $x=-6$ **(b)** $x^2-8x=0$ $x(x-8)=0$ $x=0$ or $x=8$	**(a)** $x=-1$ **or** $x=-6$ **(b)** $x=0$ **or** $x=8$
6.6	**(a)** $(x-3)(x-4)=0$ $x=3$ or $x=4$ **(b)** $x^2-5x=0$ $x(x-5)=0$ $x=0$ or $x=5$	**(a)** $x=3$ **or** $x=4$ **(b)** $x=0$ **or** $x=5$

Unit 3 · Problems with quadratics (Lesson 3.5)

Q	Working	Answer
show that, then solve		
7.1	**(a)** $(x+4)(x-2)=16$ $x^2+2x-8=16$ $x^2+2x-24=0$ **(b)** $x^2+2x-24=0$ $(x-4)(x+6)=0$ $x=4$ (a length is positive) length $=4+4=8$	**(a) shown** **(b) 8 cm**
7.2	**(a)** $(x+5)(x-1)=27$ $x^2+4x-5=27$ $x^2+4x-32=0$ **(b)** $x^2+4x-32=0$ $(x-4)(x+8)=0$ $x=4$ (a length is positive) length $=4+5=9$	**(a) shown** **(b) 9 cm**

7.3	**(a)** $(x+6)(x+1)=50$ $x^2+7x+6=50$ $x^2+7x-44=0$ **(b)** $x^2+7x-44=0$ $(x-4)(x+11)=0$ $x=4$ (a length is positive) $\text{width}=4+1=5$	**(a) shown** **(b) 5 cm**
7.4	**(a)** $(x+3)(x-2)=24$ $x^2+x-6=24$ $x^2+x-30=0$ **(b)** $x^2+x-30=0$ $(x-5)(x+6)=0$ $x=5$ (a length is positive) $\text{length}=5+3=8$	**(a) shown** **(b) 8 cm**
7.5	**(a)** $(x+7)(x-3)=24$ $x^2+4x-21=24$ $x^2+4x-45=0$ **(b)** $x^2+4x-45=0$ $(x-5)(x+9)=0$ $x=5$ (a length is positive) $\text{width}=5-3=2$	**(a) shown** **(b) 2 cm**
7.6	**(a)** $(x+2)(x+1)=56$ $x^2+3x+2=56$ $x^2+3x-54=0$ **(b)** $x^2+3x-54=0$ $(x-6)(x+9)=0$ $x=6$ (a length is positive) $\text{length}=6+2=8$	**(a) shown** **(b) 8 cm**

Unit 3 · Perfect squares (Lesson 3.4)

Q	Working	Answer
is it a perfect square?		
8.1	$10 \div 2 = 5,\ 5^2 = 25$	**Yes,** $(x+5)^2$
8.2	$-8 \div 2 = -4,\ (-4)^2 = 16$	**Yes,** $(x-4)^2$
8.3	$6 \div 2 = 3,\ 3^2 = 9 \neq 8$ $x^2 + 6x + 8 = (x+2)(x+4)$	**No**
8.4	$-14 \div 2 = -7,\ (-7)^2 = 49$	**Yes,** $(x-7)^2$
8.5	$12 \div 2 = 6,\ 6^2 = 36 \neq 30$	**No**
8.6	$-2 \div 2 = -1,\ (-1)^2 = 1$	**Yes,** $(x-1)^2$